



JABchem



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Past Papers Higher Chemistry

2016 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/120)	%	
A	86+	71.7%	29.6%
B	74+	61.7%	25.2%
C	62+	51.7%	21.3%
D	56+	46.7%	8.8%
No award	<56	<46.7%	15.1%

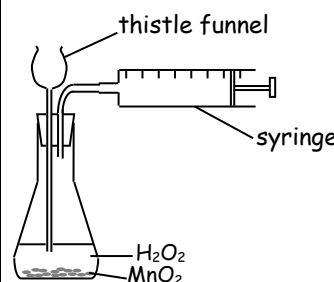
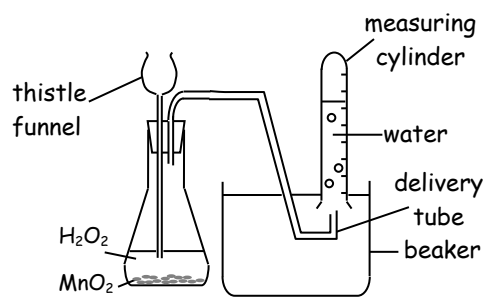
Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	13.0 /20	48.5 /80	13.2 /20

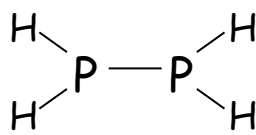
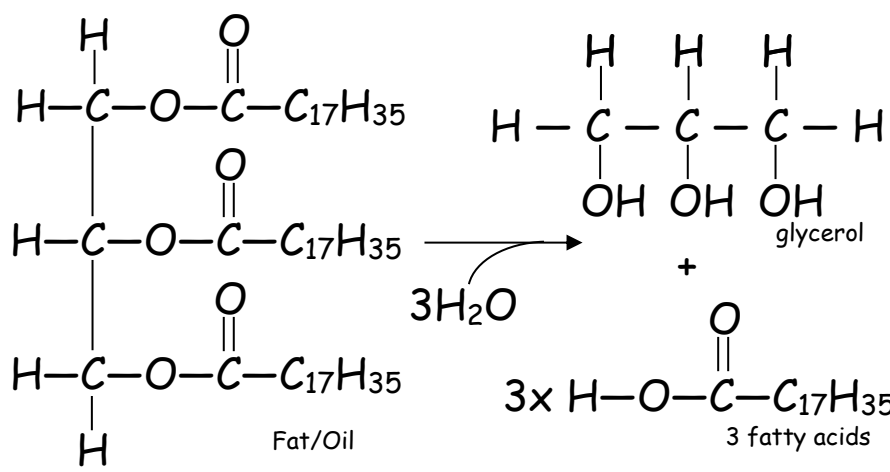
2016 Higher Chemistry Marking Scheme

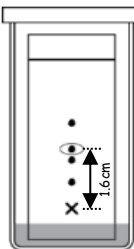
MC Qu	Answer	% Pupils Correct	Reasoning
1	D	55	☒ A $(\text{Na}^+)_2\text{S}^{2-}$ electron arrangements: Na^+ 2,8 and S^{2-} 2,8,8 ☒ B $\text{Mg}^{2+}(\text{Cl}^-)_2$ electron arrangements: Mg^{2+} 2,8 and Cl^- 2,8,8 ☒ C K^+Br^- electron arrangements: K^+ 2,8,8 and Br^- 2,8,18,8 ☒ D $\text{Ca}^{2+}(\text{Cl}^-)_2$ electron arrangements: Ca^{2+} 2,8,8 and Cl^- 2,8,8
2	A	80	☒ A H has the δ^+ charge and Cl has the δ^- charge as it has higher electronegativity ☒ B Electrons will be closer to the more electronegative element (H 2.2 and Cl 3.0) ☒ C Chlorine is the more electronegative element and has the δ^- charge ☒ D Electrons will be closer to the more electronegative element (H 2.2 and Cl 3.0)
3	A	83	☒ A Electronegativity difference $\text{CsF}_2 = 4.0 - 0.8 = 3.2 \therefore$ greatest ionic character ☒ B Electronegativity difference $\text{CsI}_2 = 2.6 - 0.8 = 1.8$ ☒ C Electronegativity difference $\text{NaF} = 4.0 - 0.9 = 3.1$ ☒ D Electronegativity difference $\text{NaI} = 2.6 - 0.9 = 1.7 \therefore$ least ionic character
4	B	81	Enthalpy change is the same for catalysed and uncatalysed reactions. Enthalpy change is measure from R to P = $50 - 100 = -50 \text{ kJ mol}^{-1}$ Enthalpy change must be negative as diagram is exothermic type (downhill overall)
5	B	93	Formula of limonene molecule = $\text{C}_{10}\text{H}_{16}$ Each isoprene unit contains 5 carbons $\therefore$ 2 isoprene units combined to make limonene
6	A	66	☒ A molecules has a lack of polar groups so stays in food when boiled in polar water ☒ B molecule has a polar -OH hydroxyl group and likely to enter boiling water ☒ C molecule has a polar -OH hydroxyl group and likely to enter boiling water ☒ D molecule has a polar -COOH carboxyl group and likely to enter boiling water
7	C	75	☒ A Hydrolysis: vegetable oil would hydrolyses into glycerol and three fatty acids ☒ B Condensation: vegetable oil made by condensation from glycerol & 3 fatty acids ☒ C Hydrogenation: hydrogen added across $\text{C}=\text{C}$ bonds making saturated chains ☒ D Dehydrogenation: removing hydrogen would make more $\text{C}=\text{C}$ bonds (stays oil)
8	D	71	☒ A reaction rate falls at temperature above 37°C ☒ B reaction rate rises not falls at temperatures from 0°C to 37°C ☒ C reaction rate falls at temperature above 37°C ☒ D Enzymes have an optimum temperature and decrease in activity as temp rises
9	C	81	☒ A Fat is a solid triester molecule made from condensation of glycerol and 3 fatty acids ☒ B Oil is a liquid triester molecule made from condensation of glycerol and 3 fatty acids ☒ C Soap is made from neutralisation of a fatty acid with an alkali ☒ D Glycerol (propane-1,2,3-triol) is a trihydric alcohol with three -OH groups
10	B	40	☒ A the emulsifier produced and the edible oil are both esters to start with. ☒ B glycerol reacts with edible oil by taking a fatty acid chain from the oil. ☒ C fatty acids are produced in the reaction of making an emulsifier ☒ D amino acids are the building blocks of making protein
11	B	61	☒ A $\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$ is the reduction reaction in Benedict's Solution ☒ B $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}^0$ is the reduction reaction in Tollen's Reagent ☒ C $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ is the reduction reaction in acidified dichromate ☒ D $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ is the reduction reaction in acidified permanganate
12	C	80	☒ A numbering system used does not give functional group ($\text{C}=\text{O}$) the lowest number ☒ B wrong functional group: -al is aldehyde group ($-\text{CHO}$) not a ketone group ☒ C 5 carbon ketone with functional group on C_2 and methyl groups on C_3 and C_4 ☒ D wrong functional group: -al is aldehyde group ($-\text{CHO}$) not a ketone group

13	C	31	Available nitric acid: no. of mol = volume × concentration = 0.2litres × 0.1mol l⁻¹ = 0.02mol $\begin{array}{ccccccc} \text{CaCO}_3 & + & 2\text{HNO}_2 & \longrightarrow & \text{Ca(NO}_3)_2 & + & \text{H}_2\text{O} + \text{CO}_2 \\ 1\text{mol} & & 2\text{mol} & & 1\text{mol} & & 1\text{mol} & & 1\text{mol} \\ 0.01\text{mol} & & 0.02\text{mol} & & 0.01\text{mol} & & 0.01\text{mol} & & 0.01\text{mol} \end{array}$ gfm CaCO₃ = (1×40.1)+(1×12)+(3×16) = 40.1+12+48 = 100.1g ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{2}{100.1} = 0.020\text{mol}$ gfm Ca(NO₃)₂ = (1×40.1)+(2×14)+(6×16) = 40.1+28+96 = 164.1g ☒ A CaCO₃ is in excess as 0.01mol reacts and more CaCO₃ is available than 0.01mol. ☒ B All nitric acid is used up in the reaction as nitric acid is the limiting reactant ☒ C 0.01mol Ca(NO₃)₂ produced ∴ mass = no. of mol × gfm = 0.01 × 164.1 = 1.64g ☒ D Volume = no. of mol × Molar Volume = 0.01mol × 24 litres mol⁻¹ = 0.24litres = 240cm³
14	B	51	☒ A carbon must be in the gaseous state in this reaction ☒ B 4mol C-F bonds are broken ∴ total enthalpy change = 4×484 = +1936kJ mol ⁻¹ ☒ C 4mol C-F bonds broken but 2mol F-F bonds formed ☒ D carbon must be in the gaseous state in this reaction
15	C	66	☒ A At equilibrium, rate of forward reaction = rate of reverse reaction ☒ B At equilibrium, concentration of reactants are constant (but not equal) ☒ C At equilibrium, concentration of reactants are constant ☒ D Activation Energies for forward and reverse reactions are never equal
16	A	75	☒ A enthalpy for the complete combustion of 1mol of a substance ☒ B Enthalpy of combustion must be complete combustion (no carbon monoxide is formed) ☒ C Enthalpy of combustion must be complete combustion (water not hydrogen is formed) ☒ D Enthalpy of combustion must be complete combustion (no carbon monoxide is formed)
17	C	68	☒ A oxidising agent is reduced itself not oxidised ☒ B oxidising agent is reduced itself not oxidised ☒ C oxidising agent is reduced itself ∴ oxidising agent gains electrons ☒ D oxidising agent is reduced itself ∴ oxidising agent gains electrons
18	D	49	Step 1: Write down main species in reaction $\text{ClO}_3^- \longrightarrow \text{Cl}_2$ Step 2: Balance all atoms other than O or H $2\text{ClO}_3^- \longrightarrow \text{Cl}_2$ Step 3: Balance O atoms by adding H₂O to the other side $2\text{ClO}_3^- \longrightarrow \text{Cl}_2 + 6\text{H}_2\text{O}$ Step 4: Balance H atoms by adding H⁺ to the other side $2\text{ClO}_3^- + 12\text{H}^+ \longrightarrow \text{Cl}_2 + 6\text{H}_2\text{O}$ Step 5: Balance charge by adding electrons to the most positive side $2\text{ClO}_3^- + 12\text{H}^+ + 10\text{e}^- \longrightarrow \text{Cl}_2 + 6\text{H}_2\text{O}$
19	D	41	2I⁻ → I₂ + 2e⁻ is an oxidation reaction (reverse of reduction reaction in data booklet) Reduction reaction must be below this reaction in the data booklet to proceed ☒ A SO₄²⁻ + 2H⁺ + 2e⁻ → SO₃²⁻ + H₂O (reduction reaction is above iodine in data booklet) ☒ B SO₃²⁻ + H₂O → SO₄²⁻ + 2H⁺ + 2e⁻ (not a reduction reaction) ☒ C 2Cr³⁺ + 7H₂O → Cr₂O₇²⁻ + 14H⁺ + 6e⁻ (not a reduction reaction) ☒ D Cr₂O₇²⁻ + 14H⁺ + 6e⁻ → 2Cr³⁺ + 7H₂O (reduction reaction & below iodine in data booklet)
20	A	54	☒ A white tile underneath helps to precisely determine the colour change at the end-point ☒ B using bottom of meniscus improves the reading of volume in burette at end-point. ☒ C repeating titrations improves reliability but not ability to precisely determine end-point ☒ D carrying out rough titration allows the region of the end-point to be worked out

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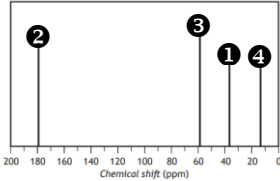
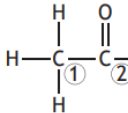
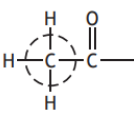
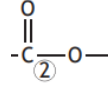
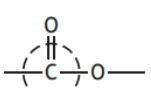
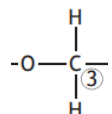
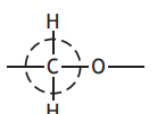
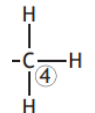
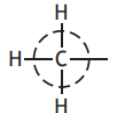
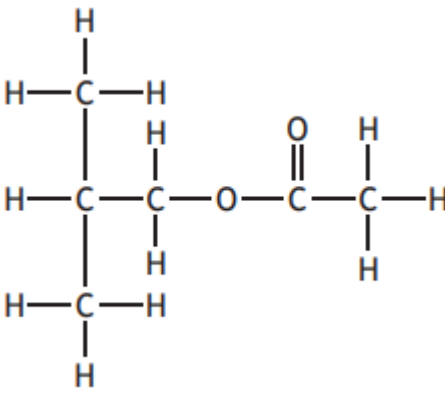
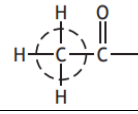
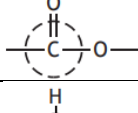
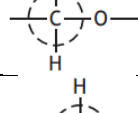
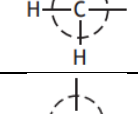
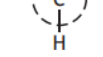
Long Qu	Answer	Reasoning																																			
1a	Increased successful collisions at higher temp	At a higher temperature, the average kinetic energy of particles increases. Number of collisions will increase and the number of collisions with energy greater than the activation energy will increase, increasing the reaction rate.																																			
1b(i)	Answer to include one of:	 																																			
1b(ii)	3.7	$\begin{aligned} \text{volume of oxygen produced} &= \text{volume strength} \times \text{volume of hydrogen peroxide solution} \\ 74\text{cm}^3 &= \text{volume strength} \times 20\text{cm}^3 \\ \text{volume strength} &= \frac{\text{volume of oxygen produced}}{\text{volume of hydrogen peroxide}} = \frac{74\text{cm}^3}{20\text{cm}^3} = 3.7 \end{aligned}$																																			
1c	80s	From graph: when concentration = 0.6mol l^{-1} the relative rate = 0.0125 s^{-1} $\text{time} = \frac{1}{\text{relative rate}} = \frac{1}{0.0125\text{ s}^{-1}} = 80\text{s}$																																			
2a(i)	Nuclear charge increases or number of protons in nucleus increases	As the nuclear charge increases (due to additional protons in the nucleus) the same outer shell of electrons undergoes a greater pull from the nucleus which causes the covalent radius/atomic size to decrease.																																			
2a(ii)	Answer to include:	1st Mark: sodium atom loses electron to become sodium ion 2nd Mark: sodium ion has two electron shells (2,8) but sodium atom has three electron shells (2,8,1)																																			
2b(i)	One answer from:	Outer electrons less strongly attracted to nucleus Outer electrons more shielded from nuclear pull as atom increases in size																																			
2b(ii)	Answer to include:	1st Mark: electron removed is being removed from a full outer shell or from an electron shell which is closer to nucleus 2nd Mark: electron being removed by 2nd ionisation energy is more strongly attracted to nucleus (or less shielded)																																			
2c(i)	Any value between 720-770	<table><tr><td>Ion in Chloride</td><td>Li⁺</td><td>Na⁺</td><td>K⁺</td><td>Rb⁺</td></tr><tr><td>Lattice Enthalpy</td><td>834</td><td>769</td><td>701</td><td>658</td></tr><tr><td>Difference</td><td></td><td>65</td><td>68</td><td>43</td></tr></table> K⁺ → Rb⁺ difference is less than previous differences <table><tr><td>Ion in Fluoride</td><td>Li⁺</td><td>Na⁺</td><td>K⁺</td><td>Rb⁺</td></tr><tr><td>Lattice Enthalpy</td><td>1030</td><td>910</td><td>808</td><td>-</td></tr><tr><td>Difference</td><td></td><td>120</td><td>102</td><td>65</td></tr><tr><td>Prediction</td><td>-</td><td>-</td><td>-</td><td>743</td></tr></table>	Ion in Chloride	Li ⁺	Na ⁺	K ⁺	Rb ⁺	Lattice Enthalpy	834	769	701	658	Difference		65	68	43	Ion in Fluoride	Li ⁺	Na ⁺	K ⁺	Rb ⁺	Lattice Enthalpy	1030	910	808	-	Difference		120	102	65	Prediction	-	-	-	743
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2c(ii)	As ionic radii increases the lattice enthalpy decreases	Going down group 1 (Li ⁺ → Na ⁺ → K ⁺ → Rb ⁺) ionic radius increases and for both fluorides and chloride the lattice enthalpy decreases. The corresponding lattice enthalpies for each chloride and fluoride show that the chloride has a lower lattice enthalpy than the fluoride for the same metal ion.																																			

3a	Covalent molecular	Phosphine PH ₃ contains only non-metal atoms in the compound and will have covalent bonding as a result. PH ₃ will have a molecular structure as it is a gas at room temperature.												
3b	1200	$1\text{mol AlP} = (1 \times 27) + (1 \times 31) = 27 + 31 = 58\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{2900\text{g}}{58\text{ g mol}^{-1}} = 50\text{mol}$ $\begin{array}{ccc} \text{AlP} + 3\text{H}_2\text{O} & \longrightarrow & \text{PH}_3 + \text{Al(OH)}_3 \\ \begin{array}{c} 1\text{mol} \\ 50\text{mol} \end{array} & & \begin{array}{c} 1\text{mol} \\ 50\text{mol} \end{array} \\ \text{Volume} = \text{no. of mol} \times \text{Molar Volume} = 50\text{mol} \times 24\text{ litres mol}^{-1} = 1200\text{ litres} \end{array}$												
3c		Phosphorus is in group 5, has a valency of 3 and makes three bonds. Hydrogen is in group 1, has a valency of 1 and makes one bond.												
4	Open Question Answer to Include:	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.						
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5a	glycerol or propane-1,2,3-triol													
5b(i)	polyunsaturated	general formula of saturated fatty acids = C_nH_{2n+1}COOH ∴ saturated 18 carbon fatty acid will have formula C₁₇H₃₅COOH <table><tr><th>No C=C bonds</th><th>1x C=C bond</th><th>2x C=C bond</th><th>3x C=C bond</th></tr><tr><td>C₁₇H₃₅COOH</td><td>C₁₇H₃₃COOH</td><td>C₁₇H₃₁COOH</td><td>C₁₇H₂₉COOH</td></tr><tr><td>Saturated</td><td>Monosaturated</td><td>Polyunsaturated</td><td>Polysaturated</td></tr></table>	No C=C bonds	1x C=C bond	2x C=C bond	3x C=C bond	C ₁₇ H ₃₅ COOH	C ₁₇ H ₃₃ COOH	C ₁₇ H ₃₁ COOH	C ₁₇ H ₂₉ COOH	Saturated	Monosaturated	Polyunsaturated	Polysaturated
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C ₁₇ H ₃₅ COOH	C ₁₇ H ₃₃ COOH	C ₁₇ H ₃₁ COOH	C ₁₇ H ₂₉ COOH											
Saturated	Monosaturated	Polyunsaturated	Polysaturated											
5b(ii)	Octanoic acid	Caprylic acid contains 8 carbons, a carboxyl -COOH group and no C=C double bonds.												
5c(i)	Answer to include:	<u>1st Mark:</u> Bromine solution is added to both until the bromine is no longer decolourised. (or reddish-brown colour remains) <u>2nd Mark:</u> more bromine would be added to olive oil as it is more unsaturated than coconut oil												
5c(ii)	Hexane is non-polar or water is polar	As oils are non-polar substances which do not dissolve in water ∴ water (a polar solvent) is of no use as a solvent for oil. Hexane is a hydrocarbon, is non-polar and will dissolve oil												
5c(iii)	Answer to include:	<u>1st Mark:</u> Coconut oil has more straight chains due to fewer double bonds while olive oil has more kinked chains due to more double bonds <u>2nd Mark:</u> London dispersion forces between coconut oil are stronger than in olive oil												

6a	hexapeptide	<table><tr><td>Prefix</td><td>Mono-</td><td>di-</td><td>Tri-</td><td>Tetra-</td><td>Penta-</td><td>Hexa-</td><td>Poly-</td></tr><tr><td>Meaning</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>many</td></tr></table>	Prefix	Mono-	di-	Tri-	Tetra-	Penta-	Hexa-	Poly-	Meaning	1	2	3	4	5	6	many				
Prefix	Mono-	di-	Tri-	Tetra-	Penta-	Hexa-	Poly-															
Meaning	1	2	3	4	5	6	many															
6b	Answer showing:	<table><tr><td>isoleucine</td><td>leucine</td><td>glycine</td><td>valine</td><td>serine</td></tr></table> Overlapping fragments: <table><tr><td>isoleucine</td><td>leucine</td><td></td><td></td><td></td></tr><tr><td></td><td>leucine</td><td>glycine</td><td>valine</td><td></td></tr><tr><td></td><td></td><td>glycine</td><td>valine</td><td>serine</td></tr></table>	isoleucine	leucine	glycine	valine	serine	isoleucine	leucine					leucine	glycine	valine				glycine	valine	serine
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isoleucine	leucine																					
	leucine	glycine	valine																			
		glycine	valine	serine																		
6c	essential	Amino acids which must be obtained by the body through the diet are known as essential amino acids.																				
6d(i)	One answer from:	<table><tr><td>Peptide only contained four different amino acids</td><td>An amino acid is repeated in the sequence</td><td>Two amino acids had the same R_f value (for that solvent)</td><td>Two different amino acids moved the same distance</td></tr></table>	Peptide only contained four different amino acids	An amino acid is repeated in the sequence	Two amino acids had the same R _f value (for that solvent)	Two different amino acids moved the same distance																
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6d(ii)	2 nd Spot from Top circled	Measured from the diagram: solvent front moved: 4cm From the Question: R_f value of methionine: 0.40 $R_f = \frac{\text{Distance moved by the substance}}{\text{Maximum distance moved by the solvent}}$ ∴ Distance moved by the Substance = R_f × Maximum distance moved by the solvent = 0.40 × 4cm = 1.6cm 																				
6e(i)	One from:	<chem>NC(C)C(=O)O</chem><chem>CC(C)C(=O)O</chem>																				
6e(ii)	30g	<table><tr><td>1kg adult</td><td>↔</td><td>100mg lethal dose alpha-amanitin for humans</td></tr><tr><td>75kg adult</td><td>↔</td><td>100mg lethal dose alpha-amanitin for humans × ⁷⁵/₁ = 7500mg alpha-amanitin</td></tr><tr><td>250mg alpha-amanitin</td><td>↔</td><td>1.0g death cap mushroom</td></tr><tr><td>7500mg alpha-amanitin</td><td>↔</td><td>1.0g death cap mushroom × ⁷⁵⁰⁰/₂₅₀ = 30g death cap mushroom</td></tr></table>	1kg adult	↔	100mg lethal dose alpha-amanitin for humans	75kg adult	↔	100mg lethal dose alpha-amanitin for humans × ⁷⁵ / ₁ = 7500mg alpha-amanitin	250mg alpha-amanitin	↔	1.0g death cap mushroom	7500mg alpha-amanitin	↔	1.0g death cap mushroom × ⁷⁵⁰⁰ / ₂₅₀ = 30g death cap mushroom								
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7a	One answer from:	<table><tr><td>UV light is damaging/harmful to skin</td><td>Sunblocks contain free-radical scavengers</td><td>UV light damages collagen</td></tr><tr><td>UV light can break bonds/molecules in skin</td><td>UV light can cause skin cancer</td><td>UV light ages skin</td></tr><tr><td>UV light creates free radicals or initiates free-radical chain reactions</td><td>UV light causes photo ageing</td><td>UV light causes sunburn</td></tr></table>	UV light is damaging/harmful to skin	Sunblocks contain free-radical scavengers	UV light damages collagen	UV light can break bonds/molecules in skin	UV light can cause skin cancer	UV light ages skin	UV light creates free radicals or initiates free-radical chain reactions	UV light causes photo ageing	UV light causes sunburn											
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UV light creates free radicals or initiates free-radical chain reactions	UV light causes photo ageing	UV light causes sunburn																				
7b(i)	Atom/molecule with unpaired electron	Free radical are species (atoms or molecules) which have an unpaired electron in the structure. This electron is very reactive and will react with a variety of other chemicals to pair up this unpaired electron.																				
7b(ii)	Initiation	<table><tr><td>Step</td><td>Reactants (before Arrow)</td><td>→</td><td>Products (after Arrow)</td></tr><tr><td>Initiation</td><td>No free radicals on Left Hand Side</td><td>→</td><td>Free radicals on Right Hand Side</td></tr><tr><td>Propagation</td><td colspan="3">Free Radicals found on both sides of arrow</td></tr><tr><td>Termination</td><td>Free radicals on Left Hand Side</td><td>→</td><td>No free radicals on Right Hand Side</td></tr></table>	Step	Reactants (before Arrow)	→	Products (after Arrow)	Initiation	No free radicals on Left Hand Side	→	Free radicals on Right Hand Side	Propagation	Free Radicals found on both sides of arrow			Termination	Free radicals on Left Hand Side	→	No free radicals on Right Hand Side				
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7b(iii)	Carboxyl group	<table><tr><td><chem>-O-H</chem></td><td><chem>-C(=O)-OH</chem></td><td><chem>-C(=O)-H</chem></td><td><chem>-C(=O)-C-</chem></td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td><td>aldehyde group</td><td>ketone group</td></tr></table>	<chem>-O-H</chem>	<chem>-C(=O)-OH</chem>	<chem>-C(=O)-H</chem>	<chem>-C(=O)-C-</chem>	hydroxyl group	carboxyl group	aldehyde group	ketone group												
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hydroxyl group	carboxyl group	aldehyde group	ketone group																			

7b(iv)	2 carbons attached to the Carbon with the -OH group	<table><tr><th>Alcohol</th><th>Description</th></tr><tr><td>Primary</td><td>1 carbon directly attached to the carbon with the -OH group</td></tr><tr><td>Secondary</td><td>2 carbons directly attached to the carbon with the -OH group</td></tr><tr><td>Tertiary</td><td>3 carbons directly attached to the carbon with the -OH group</td></tr></table>	Alcohol	Description	Primary	1 carbon directly attached to the carbon with the -OH group	Secondary	2 carbons directly attached to the carbon with the -OH group	Tertiary	3 carbons directly attached to the carbon with the -OH group
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Secondary	2 carbons directly attached to the carbon with the -OH group									
Tertiary	3 carbons directly attached to the carbon with the -OH group									
7c(i)	Diagram showing:	$\begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{O} & \text{H} \\ & & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{O} & - \text{C} & - \text{C} & - \text{H} \\ & & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & & \text{H} \end{array}$								
7c(ii)	To condense vapours which would evaporate	Both the reactants and the products in an esterification reaction are volatile and evaporate when in a hot water bath. The wet paper towel attached by an elastic band provides a cool glass surface in the inside of the test tube which will condense the vapours back to liquids and prevent their escape. These vapours are flammable.								
7c(iii)	60.7%	1mol ethanol C₂H₅OH = 46g								

9b	-414	① $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$ $\Delta\text{H} = -286 \text{ kJ mol}^{-1}$ ② $\text{Ca} + \text{O}_2 + \text{H}_2 \rightarrow \text{Ca(OH)}_2$ $\Delta\text{H} = -986 \text{ kJ mol}^{-1}$ ①x-2 $2\text{H}_2\text{O} + \rightarrow 2\text{H}_2 + \text{O}_2$ $\Delta\text{H} = +572 \text{ kJ mol}^{-1}$ ② $\text{Ca} + \text{O}_2 + \text{H}_2 \rightarrow \text{Ca(OH)}_2$ $\Delta\text{H} = -986 \text{ kJ mol}^{-1}$ Add ①'+② $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2$ $\Delta\text{H} = -414 \text{ kJ mol}^{-1}$																											
10	Open Question Answer to Include:	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.																					
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11a(i)	A diagram showing:	Flask with long narrow neck and a single gradation mark which goes completely across (or is labelled) on the narrow neck.																											
11a(ii)	Answer to include:	<table><tr><td>1st Mark:</td><td>Accurate method for volume measurement e.g. uses a pipette or burette (syringe not accepted)</td></tr><tr><td>2nd Mark:</td><td>Description of weighing by difference (or used Tare on a balance)</td></tr></table>	1 st Mark:	Accurate method for volume measurement e.g. uses a pipette or burette (syringe not accepted)	2 nd Mark:	Description of weighing by difference (or used Tare on a balance)																							
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11a(iii)	Line graph to include:	<table><tr><td>1st Mark:</td><td>All points plotted correctly ($\pm \frac{1}{2}$ box tolerance)</td></tr><tr><td>2nd Mark:</td><td>Best Fit Line</td></tr></table>	1 st Mark:	All points plotted correctly ($\pm \frac{1}{2}$ box tolerance)	2 nd Mark:	Best Fit Line																							
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2 nd Mark:	Best Fit Line																												
11a(iv) Part A	Dissolved gas/bubbles will affect the density/mass/volume	The gas bubbles in the soft drink will alter the volume and the mass of the liquid being measured which will change the density.																											
11a(iv) Part B	3.43%	$\begin{array}{l} \text{Density of sugar in g cm}^{-3} = (0.0204 \times \% \text{concentration of sugars in solution}) + 1.00 \\ 1.07 = (0.0204 \times \% \text{concentration of sugars in solution}) + 1.00 \\ 1.07 - 1.00 = (0.0204 \times \% \text{concentration of sugars in solution}) \\ 0.07 = 0.0204 \times \% \text{concentration of sugars in solution} \\ \frac{0.07}{0.0204} = \% \text{concentration of sugars in solution} \\ 3.43\text{g cm}^{-3} = \% \text{concentration of sugars in solution} \end{array}$																											
11a(v)	34.98g	$\begin{array}{l} 100\text{cm}^3 \text{ contains } 10.6\text{g sugar} \\ 330\text{cm}^3 \text{ contains } 10.6\text{g} \times \frac{330}{100} \\ = 34.98\text{g} \end{array}$																											
11b(i)		<table><tr><td></td><td></td><td></td><td></td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td><td>aldehyde group</td><td>ketone group</td></tr></table>					hydroxyl group	carboxyl group	aldehyde group	ketone group																			
hydroxyl group	carboxyl group	aldehyde group	ketone group																										
11b(ii)	$\begin{array}{c} \text{C}_6\text{H}_{12}\text{O}_6 + \text{H}_2\text{O} \\ \downarrow \\ \text{C}_6\text{H}_{12}\text{O}_7 + 2\text{H}^+ + 2\text{e}^- \end{array}$	$\begin{array}{l} \text{Redox} \quad \text{C}_6\text{H}_{12}\text{O}_6 + 2\text{Cu}^{2+} + \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_7 + 2\text{Cu}^+ + 2\text{H}^+ \\ \text{Oxidation} \quad \text{C}_6\text{H}_{12}\text{O}_6 + \quad \quad \quad \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_7 + 2\text{e}^- + 2\text{H}^+ \\ \text{Reduction} \quad \quad \quad 2\text{Cu}^{2+} + 2\text{e}^- \rightarrow \quad \quad \quad 2\text{Cu}^+ \end{array}$																											
11b(iii)	blue to orange/brick red	The final colours accepted: orange, red, brick red, brown, yellow or green																											
11b(iv)	0.0099	$\begin{array}{l} \text{no. of mol Cu}^{2+} = \text{volume} \times \text{concentration} = 0.0198\text{litres} \times 0.0250\text{mol l}^{-1} = 0.000495\text{mol} \\ \text{C}_6\text{H}_{12}\text{O}_6 + 2\text{Cu}^{2+} + \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_7 + 2\text{Cu}^+ + 2\text{H}^+ \\ \begin{array}{cc} 1\text{mol} & 2\text{mol} \\ 0.0002475\text{mol} & 0.000495\text{mol} \end{array} \\ \text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{0.0002475\text{mol}}{0.025\text{litres}} = 0.0099 \text{ mol l}^{-1} \end{array}$																											
12a(i)	Hydrogen Bonding	Hydrogen bonds form between molecules containing -OH: <table><tr><th>Isomer</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>Type</td><td>Carboxylic Acid</td><td>Carboxylic Acid</td><td>Carboxylic Acid</td><td>Ester</td><td>Ester</td><td>Ester</td><td>Ester</td><td>Ester</td></tr><tr><td>Hydrogen Bonding</td><td>yes</td><td>yes</td><td>yes</td><td>no</td><td>no</td><td>no</td><td>no</td><td>no</td></tr></table>	Isomer	1	2	3	4	5	6	7	8	Type	Carboxylic Acid	Carboxylic Acid	Carboxylic Acid	Ester	Ester	Ester	Ester	Ester	Hydrogen Bonding	yes	yes	yes	no	no	no	no	no
Isomer	1	2	3	4	5	6	7	8																					
Type	Carboxylic Acid	Carboxylic Acid	Carboxylic Acid	Ester	Ester	Ester	Ester	Ester																					
Hydrogen Bonding	yes	yes	yes	no	no	no	no	no																					

12a(ii)	Answer to include	1 st Mark:	More branching lowers boiling point			
		2 nd Mark:	The shorter the alcohol, the lower the boiling point	The longer the carboxylic acid the lower the boiling point	The nearer the ester link is to the right hand end (of the molecule) the higher the boiling point.	
12a(iii)	Temperature between 99°C and 124°C	Structures 1, 2 and 3 show that increasing number of branches decreases the boiling point. Structures 4, 5 and 6 show that increasing number of branches decreases the boiling point. Both comparisons above show that more branches lowers the boiling point. Only structures 7 and 8 are compared with structure in question as they are isomers with same formula. Boiling point must be greater than 98°C as it has less branches than structure 8 and must be less than 126°C as it has more branches than structure 7.				
12b(i)		Peak	Structure	Chemical Environment	Chemical Shift (ppm)	
		1			20-50	
		2			170-185	
		3			50-90	
		4			10-15	
12b(ii)	5	Structure		Peak	Chemical Environment	Chemical Shift (ppm)
				1		20-50
				2		170-185
				3		50-90
				2x 4		10-15
				5		25-35

2016 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D
1	D	10	11	23	55
2	A	80	10	7	2
3	A	83	5	8	4
4	B	7	81	9	2
5	B	3	93	3	1
6	A	66	8	9	16
7	C	11	5	75	8
8	D	5	5	19	71
9	C	3	2	81	14
10	B	21	40	25	13
11	B	10	61	16	12
12	C	8	2	80	9
13	C	27	10	31	31
14	B	13	51	16	20
15	C	2	12	66	20
16	A	75	9	12	4
17	C	3	8	68	21
18	D	5	43	3	49
19	D	19	13	25	41
20	A	54	16	15	16